Atomic Foundations of Matter

Chapter 9 | Grade 9 Science (Exploration) | Important Questions with Answers

Section A: Multiple Choice Questions (10 questions)

Q1. The Law of Conservation of Mass was proposed in 1789 by:

(a) John Dalton
(b) Joseph Proust
(c) Antoine Lavoisier
(d) Ernest Rutherford

Answer: (c) Antoine Lavoisier.

Q2. Water from any source always contains hydrogen and oxygen in a mass ratio of:

(a) 1 : 2
(b) 1 : 8
(c) 2 : 1
(d) 8 : 1

Answer: (b) 1 : 8. So 9 g of water always gives 1 g of hydrogen and 8 g of oxygen.

Q3. Which of the following is NOT a postulate of Dalton's atomic theory?

(a) All matter is made of very tiny particles called atoms
(b) Atoms can be created or destroyed in a chemical reaction
(c) Atoms of different elements have different masses and properties
(d) Atoms combine in the ratio of simple whole numbers to form compounds

Answer: (b). According to Dalton, atoms are indivisible and cannot be created or destroyed in a chemical reaction; they only rearrange.

Q4. Two oxygen atoms in an O_2 molecule are joined by a:

(a) Single covalent bond
(b) Double covalent bond
(c) Triple covalent bond
(d) Ionic bond

Answer: (b) Double covalent bond. Each oxygen atom has 6 valence electrons and shares 2 electrons with the other.

Q5. The ionic bond is the:

(a) Sharing of electron pairs between atoms
(b) Force of attraction between neutrons and protons
(c) Electrostatic force of attraction between oppositely charged ions
(d) Force of repulsion between like charges

Answer: (c) Electrostatic force of attraction between oppositely charged ions.

Q6. The chemical formula of aluminium oxide is:

(a) AlO
(b) Al_2O_3
(c) Al_3O_2
(d) AlO_3

Answer: (b) Al_2O_3. Criss-cross the charges of Al^{3+} and O^{2-}.

Q7. The molecular mass of nitric acid (HNO_3) is (H = 1 u, N = 14 u, O = 16 u):

(a) 31 u
(b) 47 u
(c) 63 u
(d) 79 u

Answer: (c) 63 u. 1 + 14 + (16 x 3) = 63 u.

Q8. The formula unit mass of calcium nitrate, $Ca(NO_3)_2$, is (Ca = 40 u, N = 14 u, O = 16 u):

(a) 102 u
(b) 164 u
(c) 150 u
(d) 132 u

Answer: (b) 164 u. 40 + 2 x (14 + 48) = 40 + 124 = 164 u.

Q9. N_2O_4 is correctly named as:

(a) Nitrogen oxide
(b) Dinitrogen tetraoxide
(c) Dinitrogen tetroxide
(d) Nitrogen tetroxide

Answer: (c) Dinitrogen tetroxide. The last vowel of the prefix 'tetra' is dropped before 'oxide'.

Q10. Which statement is true for ionic compounds like sodium chloride?

(a) They have low melting points
(b) They conduct electricity in the solid state
(c) They are soluble in water and conduct electricity when dissolved
(d) They are soluble in kerosene and petrol

Answer: (c). Ions become free to move only when dissolved in water (or molten). In the solid state ions are fixed, so there is no conduction.

Section B: True or False (5 questions)

Q11. There is practically no change in mass when salt dissolves in water.

Answer: True. Dissolving is a physical change, and the mass of the solution equals the sum of the masses of salt and water.

Q12. In Activity 9.2 (open conical flask), the final reading equals the initial reading because mass is conserved.

Answer: False. In the open flask the carbon dioxide gas escapes, so the final reading is lower. In the closed set-up (balloon fixed to the flask) the readings match.

Q13. The Law of Definite Proportions is true for mixtures as well as compounds.

Answer: False. It holds only for compounds, where elements combine in a fixed ratio by mass. In a mixture the components can be present in any ratio.

Q14. Ionic compounds do not exist as single molecules; they form three-dimensional crystals.

Answer: True. For example, in NaCl each Na^+ is surrounded by six Cl^- ions and vice versa, in a repeating crystal lattice.

Q15. Metals generally form anions by gaining electrons.

Answer: False. Metals generally lose electrons to form cations (e.g. Na^+, Mg^{2+}). Non-metals generally form anions.

Section C: Fill in the Blanks (5 questions)

Q16. The force that holds atoms together in a molecule or compound is called a _____.

Answer: chemical bond.

Q17. The Law of Constant Proportions is also called the Law of _____ Proportions or Proust's Law.

Answer: Definite.

Q18. A positively charged ion is called a _____ and a negatively charged ion is called an _____.

Answer: cation; anion.

Q19. The formula unit mass of sodium oxide (Na_2O) is _____ u (Na = 23 u, O = 16 u).

Answer: 62 u. (23 x 2) + 16 = 62 u.

Q20. A covalent bond is formed by the _____ of electrons between atoms, while an ionic bond is formed by the _____ of electrons.

Answer: sharing; transfer.

Section D: Assertion and Reason (4 questions)

Q21. Assertion (A): In a closed container, the total mass of the products equals the total mass of the reactants.
Reason (R): Atoms are neither created nor destroyed in a chemical reaction; they only rearrange.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (i) Both are true and R explains A. Since no atoms are gained or lost, the total mass stays the same.

Q22. Assertion (A): Neon neither transfers nor shares its valence electrons.
Reason (R): Neon has a complete octet (2, 8) in its outermost shell, so it is already stable.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (i) Both are true and R explains A. A complete octet makes the atom stable and unreactive.

Q23. Assertion (A): Solid sodium chloride does not conduct electricity, but its aqueous solution does.
Reason (R): In the solid state the ions are held in fixed positions, whereas in water they are free to move.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (i) Both are true and R explains A. Conduction needs free-moving ions.

Q24. Assertion (A): A solution of sugar in water does not conduct electricity.
Reason (R): Sugar is an ionic compound.

(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

Answer: (iii) A is true, but R is false. Sugar is a covalent compound. It dissolves in water but does not give ions in solution, so the solution does not conduct electricity.

Section E: Short Answer Questions (6 questions)

Q25. State the Law of Conservation of Mass. Why does the reading drop in the open flask when baking soda reacts with vinegar?

Answer: Matter can neither be created nor destroyed in a chemical reaction (Lavoisier, 1789); the total mass of reactants equals the total mass of products. In the open flask, the carbon dioxide gas formed escapes into the air, so the final reading is lower. If the gas is trapped (balloon fixed on the flask), the readings are the same.

Q26. State the Law of Constant Proportions. If 9 g of water is decomposed, what masses of hydrogen and oxygen are obtained?

Answer: In any compound, the elements are present in a fixed ratio by mass, irrespective of the source or method of preparation. Hydrogen : oxygen in water = 1 : 8, so 9 g of water gives 1 g of hydrogen and 8 g of oxygen.

Q27. Carbon monoxide contains carbon and oxygen in the mass ratio 3 : 4. How much oxygen will combine with 9 g of carbon?

Answer: Oxygen required = (4 / 3) x 9 g = 12 g.

Q28. 5.3 g of sodium carbonate and 6.0 g of acetic acid react to give 2.2 g of carbon dioxide, 0.9 g of water and 8.2 g of sodium acetate. Verify the Law of Conservation of Mass.

Answer: Mass of reactants = 5.3 + 6.0 = 11.3 g. Mass of products = 2.2 + 0.9 + 8.2 = 11.3 g. Since the mass of reactants equals the mass of products, the law is obeyed.

Q29. Write the chemical formulae of: (i) aluminium sulfate, (ii) magnesium hydroxide, (iii) ferric oxide, (iv) ammonium chloride.

Answer: (i) $Al_2(SO_4)_3$ (ii) $Mg(OH)_2$ (iii) Fe_2O_3 (iv) NH_4Cl. (Criss-cross the charges: Al^{3+}/SO_4^{2-}, Mg^{2+}/OH^-, Fe^{3+}/O^{2-}, NH_4^+/Cl^-.)

Q30. Find the formula unit mass of (i) magnesium hydroxide, $Mg(OH)_2$, and (ii) potassium chloride, KCl. (Mg = 24 u, O = 16 u, H = 1 u, K = 39 u, Cl = 35.5 u)

Answer: (i) $Mg(OH)_2$ = 24 + 2 x (16 + 1) = 24 + 34 = 58 u. (ii) KCl = 39 + 35.5 = 74.5 u.

Section F: Long Answer Questions (5 questions)

Q31. Describe an experiment to verify the Law of Conservation of Mass using baking soda and vinegar. What do you conclude?

Answer:

Set-up: Take a clean, dry 100 mL conical flask, pour about 20 mL of vinegar into it, and fix a balloon containing about 2 g of baking soda (sodium hydrogencarbonate) to its mouth with a thread, without letting the soda fall in. Weigh the whole set-up and note the reading.

Procedure: Lift the balloon end so that the baking soda falls into the vinegar. A brisk effervescence occurs and carbon dioxide inflates the balloon. Weigh the set-up again.

Observation: The final reading is the same as the initial reading. (In the open-flask version, the gas escapes and the reading drops.)

Conclusion: The total mass of vinegar and baking soda before the reaction equals the total mass of carbon dioxide and other substances after it. Matter is neither created nor destroyed in a chemical reaction. This is the Law of Conservation of Mass.

Q32. State the postulates of Dalton's atomic theory. How does it explain the Law of Conservation of Mass and the Law of Constant Proportions?

Answer:

Postulates (1808): (1) All matter is made of very tiny particles called atoms, which take part in chemical reactions. (2) Atoms are indivisible and cannot be created or destroyed in a chemical reaction. (3) Atoms of a given element are identical in mass and chemical properties. (4) Atoms of different elements have different masses and chemical properties. (5) Atoms combine in the ratio of simple whole numbers to form compounds. (6) The relative number and kinds of atoms are constant in a given compound.

Conservation of Mass: Atoms are not created or destroyed during a reaction; they are only rearranged. So the total mass remains the same.

Constant Proportions: Since atoms have fixed masses and combine in a fixed whole-number ratio in a given compound, the elements are always present in a fixed ratio by mass.

Q33. Explain the formation of covalent bonds with the help of hydrogen (H_2), chlorine (Cl_2), oxygen (O_2) and hydrogen chloride (HCl) molecules.

Answer:

A covalent bond is formed by the sharing of electrons between atoms so that each completes its octet (or duplet).

H_2: Each H atom has 1 electron in the K-shell and needs 1 more. Two H atoms share one pair of electrons: a single bond, H-H.

Cl_2: Each Cl atom (2, 8, 7) needs 1 electron. Two Cl atoms share one pair: a single bond, Cl-Cl.

O_2: Each O atom (2, 6) needs 2 electrons. Two O atoms share two pairs: a double bond, O=O.

HCl: H needs 1 electron and Cl needs 1 electron, so they share one pair: a single bond, H-Cl. Thus HCl is a covalent compound.

(Similarly, in water, one O atom shares one electron with each of two H atoms to form H_2O.)

Q34. Explain the formation of sodium chloride (NaCl) by the transfer of electrons. What is an ionic bond?

Answer:

Sodium (Z = 11): electronic configuration 2, 8, 1. It loses its one valence electron to get a stable octet and becomes the sodium cation, Na^+ (11 protons, 10 electrons).

Chlorine (Z = 17): configuration 2, 8, 7. It gains one electron to complete its octet and becomes the chloride anion, Cl^-.

$Na + Cl \rightarrow Na^+ + Cl^- \rightarrow NaCl$

Ionic bond: The electrostatic force of attraction between oppositely charged ions that holds them together.

Crystal structure: Ionic compounds do not exist as single molecules. In NaCl, each Na^+ is surrounded by six Cl^- ions and each Cl^- by six Na^+ ions in a repeating 3-D crystal lattice.

Q35. Differentiate between ionic and covalent compounds on the basis of formation, solubility, electrical conductivity and melting/boiling points.

Answer:

Formation: Ionic compounds are formed by transfer of electrons (usually metal + non-metal); covalent compounds are formed by sharing of electrons (usually between non-metals).

Solubility: Ionic compounds (NaCl, copper sulfate) are generally soluble in water but insoluble in kerosene and petrol. Most covalent compounds (camphor, naphthalene) are insoluble in water but dissolve in kerosene and petrol.

Conductivity: Ionic compounds do not conduct in the solid state (ions are fixed) but conduct when dissolved in water, because ions become free to move. Covalent compounds do not conduct electricity: some, like sugar, dissolve in water but give no ions; others, like camphor and naphthalene, do not dissolve in water at all.

Melting and boiling points: Ionic compounds have high melting and boiling points due to strong inter-ionic attractions; covalent compounds usually have low melting and boiling points.

Units: Covalent compounds exist as molecules (molecular mass); ionic compounds exist as 3-D crystals (formula unit mass).